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A REPORT
ON
Telco Customer Churn Prediction Using Machine Learning Classification Techniques

BY

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Abstract

Telecom operators lose a share of their customer base to churn every period, and unlike many other customer behaviour signals, the decision to leave is often invisible until cancellation occurs. This report presents a data-mining pipeline addressing one central question: Given a customer's account information, service subscriptions and billing history, can we predict whether the customer will churn? The study uses the Telco Customer Churn dataset, consisting of 7,043 records and 21 columns. After removing the customer identifier, 19 attributes were retained as predictors and one attribute was used as the class label.

Initial data exploration revealed an imbalanced target distribution, with 73.46% of customers retained and 26.54% churned, corresponding to a ratio of approximately 2.77:1. The TotalCharges attribute was converted from object to numeric format, with non-convertible values replaced by zero. After removing the customer identifier, all remaining categorical attributes were one-hot encoded, producing 30 predictor columns. The data were divided using a stratified 80:20 train-test split, resulting in 5,634 training records and 1,409 testing records. All predictors were then standardised using statistics fitted exclusively on the training data.

Five classification algorithms were evaluated using the preprocessing pipeline: Decision Tree, Support Vector Machine (SVM), Gaussian Naive Bayes, a back-propagation Multilayer Perceptron (MLP) and a k-Nearest Neighbours (KNN) classifier implemented from scratch using NumPy with $k = 15$. Hyperparameter tuning using three-fold stratified cross-validation selected a Decision Tree with maximum depth 5 and minimum split size 10, an SVM with $C = 10$ and a linear kernel and an MLP with 100 hidden units, $\alpha = 0.0001$ and a learning rate of 0.01, using early stopping.

The Decision Tree achieved the highest test accuracy at 79.42%, followed closely by the MLP at 79.28%, while the SVM achieved 78.64%. The MLP produced the highest F1-score at 60.22% and the highest ROC-AUC of 0.839 among the evaluated models. The Decision Tree achieved an F1-score of 58.21% and ROC-AUC of 0.827, while the SVM achieved an F1-score of 56.81% and ROC-AUC of 0.824. The custom KNN classifier obtained 76.93% accuracy, 55.30% F1-score and 0.808 ROC-AUC. Gaussian Naive Bayes achieved 65.58% accuracy and 57.19% F1-score, with a high recall of 86.63% but low precision of 42.69%.

Overall, the results demonstrate that the Decision Tree and MLP provided the strongest predictive performance in terms of accuracy and F1-score, respectively, among the evaluated models. The MLP also achieved the highest ROC-AUC, indicating comparatively strong discrimination between churned and retained customers. These findings highlight the potential of data-mining and machine-learning techniques to identify customers at risk of churn and support proactive customer-retention strategies.

Keywords: *Telecom Customer Churn, Data Mining, Machine Learning, Customer Churn Prediction, Classification, Support Vector Machine, Multilayer Perceptron, K-Nearest Neighbours, Decision Tree, Naive Bayes.*

1. Introduction

Customer churn is a common problem faced by telecommunication companies. It happens when a customer stops using a company's services or switches to another service provider. In today's competitive market, customers have many options, so they can easily change providers if they are not satisfied with the service, price, customer support or available plans. Losing customers can affect a company's revenue, while finding new customers can require a lot of time and money. Because of this, it is important for companies to understand why customers leave and find ways to prevent them from leaving.

With the increasing use of technology, companies collect a large amount of information about their customers. This information can be analyzed to understand customer behavior and find patterns that may indicate whether a customer is likely to leave. Machine learning is especially useful for this purpose because it can learn from previous customer data and make predictions about future customers. By predicting churn in advance, companies can take action before a customer decides to leave.

This project focuses on Customer Churn Prediction using Machine Learning using the Telco Customer Churn dataset. The dataset contains information about 7,043 customers, including their tenure, contract type, internet service, payment method, monthly charges, total charges and other personal and service-related information. It also contains a Churn attribute that tells us whether a customer has left the company or is still using its services.

First, the dataset is cleaned and prepared for analysis. Missing or inappropriate values are handled and categorical information is converted into a format that can be understood by machine learning algorithms. After that, Exploratory Data Analysis (EDA) is performed to understand the data and identify factors that may have an effect on customer churn. Various graphs and charts are used to examine relationships between churn and factors such as contract type, tenure, internet service, monthly charges and other customer characteristics.

Following data preparation and exploratory analysis, different machine learning algorithms, including Decision Tree, Naive Bayes', MLP, SVM and K-Nearest Neighbors (KNN), as presented in Table 1, are applied to predict customer churn. These algorithms represent different approaches to classification and allow their predictive capabilities to be compared. The trained models are evaluated using the performance metrics presented in Table 2. Multiple evaluation metrics are considered rather than relying only on accuracy, since the dataset contains more non-churning customers than churning customers.

The main aim of this project is to develop a reliable machine learning approach for identifying customers who are more likely to churn. The results can help telecommunications companies better understand customer behavior and take preventive measures, such as providing targeted offers, improving service quality, offering suitable plans or providing personalized incentives to customers at higher risk of leaving. Such proactive strategies can help reduce customer loss and improve customer retention.

Overall, this project shows how data preprocessing, data analysis, and machine learning can be used together to solve a real-world business problem. By studying past customer data and comparing different machine learning models, the study aims to understand why customers leave and show how machine learning can help telecom companies make better decisions and highlight the potential of machine learning to support effective customer-retention and decision-making strategies in the telecommunications sector.

Table 1: Machine learning algorithms which are used in this paper

S.N.	Machine Learning Algorithms
1	Decision Tree Classifier
2	Support Vector Machine
3	K-Nearest Neighbors
4	Naïve Bayes Classifier
5	Multilayer Perceptron

Table 2: Evaluation Metrics

S.N	Evaluation Metrics
1	Accuracy
2	Precision
3	Sensitivity
4	F1 Score
5	ROC-AUC

2. Literature Review

Previous studies about customer churn in the telecommunications industry mainly looked at customer surveys, service quality and old ways of marketing.For example Ribeiro, Barbosa, Moreira and Rodrigues [2] found that how good internet and TV services are matters for keeping customers than landline services. They also saw that when customers stay loyal customer churn goes down. Also the idea of quality from Kano, Seraku, Takahashi and Tsuji [3] shows that some service features are things people just expect to have. If these features are missing people get unhappy. This idea helps us see why things like internet service and contract type can predict customer churn even if the customers do not say they are unhappy.

As machine learning has grown researchers have used these models more to predict customer churn. Vafeiadis, Diamantaras, Sarigiannidis and Chatzisavvas [4] tested five machine learning ways on a telecommunications dataset. They found that using an SVM- model with AdaBoost got almost 97% accuracy. In another study Ahmad, Jafar and Aljoumaa [5] used call data from networks and billing info from a real company and got an AUC of 93.3%. These studies show that machine learning can find links between who a customer is and their customer churn behavior. Our own early results show the thing. We found that contract type, tenure and internet service type are important signs of customer churn.

One big problem in these studies is class imbalance. This happens when there are more customers who stay than customers who leave.. Van den Poel [6] looked at this problem in customer churn prediction. They tried things like sampling and different types of forest models to get results on imbalanced data. Idris, Rizwan and Khan [7] also did something for telecommunications. They picked features and balanced the data before using Random Forest and k-Nearest Neighbors. However our data has a split of about 73:27. This is not as bad as the 90:10 split seen in studies. So we chose to handle the imbalance during the testing phase using math metrics instead of changing the original data too much. We followed the ideas of De Caigny, Coussement and De Bock [8]. Focused on making sure our models were easy to understand.

The main algorithms we used in this study are common in research about classification and customer churn. Decision trees, like the ID3 way from Quinlan [9] and the CART way from Breiman et al. [10]. We also used the Support Vector Machine from Cortes and Vapnik [11] which identifies decision boundaries between classes and can use kernel functions to model nonlinear relationships. The nearest-neighbor rule from Cover and Hart [12] provides a simple instance-based classification approach in which the class of an observation is determined using nearby observations. Naive Bayes applies probabilistic reasoning based on the assumption of conditional independence between features [13]. However, this assumption can be restrictive for the Telco Customer Churn dataset because several service-related variables may be correlated; for example, additional services may be associated with particular types of internet service. Lastly we used perceptrons from the work of Rumelhart, Hinton and Williams [14].

For evaluation, we used kfold cross-validation from Kohavi [15] to get a better idea of how well the models work on different parts of the data. We also used stopping from Prechelt [16] to stop the neural-network model from learning too much. Since we have customers who stay than customers who leave we did not just look at accuracy. We followed the advice of Chicco and Jurman [17]. What really makes our study different is not the algorithms but the careful and open design of our pipeline. Of focusing on a single high accuracy figure we used the same preprocessing and data-splitting steps, for five different algorithms, one of which we built from scratch.Three-fold stratified cross-validation was used during hyperparameter tuning, with F1-score as the optimization metric. Final model performance was evaluated on the held-out test set using multiple evaluation metrics.

3. Materials and Methods

3.1 Dataset Description and Source

The analysis is based on a customer-level dataset of the well-known Telco Customer Churn dataset, originally distributed by IBM as part of its sample data. It is widely used for customer churn prediction and retention analysis. The dataset contains 7,043 customer records and 21 attributes. It provides information about customer demographics, subscribed telecommunications services, contract characteristics, billing information and churn status.

The dataset was obtained from Kaggle: Telco Customer Churn Dataset. Each observation represents a customer. The attributes include a customer identifier. There are variables that describe demographic characteristics and subscribed services. There are variables related to tenure and billing. There is a binary target variable that indicates whether the customer churned.

The dataset contains customer information such as gender, senior citizen status, partner and dependent status, tenure, phone and internet services, contract type, payment method, monthly charges and total charges. The Churn attribute is the target variable. The Churn attribute divides customers into two groups. The two groups are customers who churned and customers who remained. This structure makes the dataset suitable for classification. This structure makes the dataset suitable, for analysis. The exploratory analysis is aimed at identifying factors associated with customer churn. The exploratory analysis is aimed at supporting targeted customer-retention strategies.

Before performing the analysis, the dataset was examined for missing values, inconsistent data types, duplicate records and other potential quality issues. The TotalCharges attribute required particular attention because it was stored as a text field rather than a numerical variable and contained blank values for some customers. These blank entries were handled appropriately before converting the attribute into a numerical format.

The customerID attribute was treated as an identifier rather than a predictive feature because it uniquely identifies customers but does not provide meaningful behavioral information. Categorical variables were reviewed for consistency, while numerical attributes such as tenure, MonthlyCharges and TotalCharges were checked to ensure that they could be used correctly during analysis and model development.

No unnecessary records were retained after preprocessing and the cleaned dataset was subsequently used for exploratory data analysis and predictive modeling. These data-quality checks helped ensure that the results were not significantly affected by missing, incorrectly formatted or irrelevant values.

The target variable, Churn, is binary and moderately imbalanced: of the 1,409 customers in the held-out test partition, 1,036 (73.5%) did not churn and 373 (26.5%) churned, a proportion consistent with the class balance reported for this dataset in the wider literature.

Table 3: Description of attributes in the customer churn dataset

SN	Feature	Column name	Description
1	customerID	customerID	Unique customer identifier (removed prior to modeling)
2	gender	gender	Customer gender (Female / Male)
3	Senior citizen	SeniorCitizen	Whether the customer is a senior citizen (0 = No, 1 = Yes)
4	Partner	Partner	Whether the customer has a partner (Yes/No)
5	Dependents	Dependents	Whether the customer has dependents (Yes/No)
6	Tenure	tenure	Number of months the customer has stayed with the company
7	Phone service	PhoneService	Whether the customer has phone service (Yes/No)
8	Multiple lines	MultipleLines	Whether the customer has multiple lines (Yes/No/No phone service)
9	Internet service	InternetService	Type of internet service (DSL/Fiber optic/No)
10	Online security	OnlineSecurity	Whether the customer has online security add-on
11	Online backup	OnlineBackup	Whether the customer has online backup add-on
12	Device protection	DeviceProtection	Whether the customer has device protection add-on
13	Tech support	TechSupport	Whether the customer has tech support add-on
14	Streaming TV	StreamingTV	Whether the customer has streaming TV service
15	Streaming movies	StreamingMovies	Whether the customer has streaming movies service
16	Contract	Contract	Contract term (Month-to-month/One year/Two year)
17	Paperless billing	PaperlessBilling	Whether the customer uses paperless billing (Yes/No)
18	Payment method	PaymentMethod	Payment method (Electronic check, Mailed check, Bank transfer, Credit card)
19	Monthly charges	MonthlyCharges	Amount charged to the customer monthly (USD)
20	Total charges	TotalCharges	Total amount charged to the customer (USD)
21	Churn (target)	Churn	Class label: whether the customer left the company (Yes/No)

3.2 Data Exploration and Visualization

Before applying the machine learning models, the dataset was examined to understand its structure, data types and missing values using pandas' built-in profiling utilities. Using df.info(), it was confirmed that the dataset contains 7,043 rows and 21 columns. The dataset includes numerical attributes such as SeniorCitizen, tenure and MonthlyCharges, while most other attributes are categorical. The TotalCharges column is numeric in nature but was initially stored as text because some records contained blank values.The df.isnull().sum() check showed zero explicit null values across all 21 columns, although this initial check did not detect the blank-string entries later found in TotalCharges, which required a targeted conversion step described in Section 4. However, blank entries were identified in the TotalCharges column during further inspection. These values were handled during the data-preprocessing stage by converting the column into a numeric format and appropriately managing the resulting missing values.

Descriptive statistics were computed for the three purely numeric columns available at that stage of the pipeline (`SeniorCitizen`, `tenure`, and `MonthlyCharges`) using `df.describe()`. The results are summarized in Table 2. The `tenure` column ranges from 0 to 72 months, with a mean of approximately 32 months, indicating that the customer base includes both new subscribers and long-tenured customers with up to six years of service. `MonthlyCharges` ranges from $18.25 to $118.75, with a mean of $64.76, reflecting the variety of service bundles offered by the provider, from basic phone-only plans to full fiber-optic and streaming packages. Only 16.2% of customers are senior citizens, indicating that the customer base is predominantly composed of non-senior customers.

Table 4 Descriptive statistics of the numeric features (prior to preprocessing)

Feature	Mean	Std	Min	25%	50%	75%	Max
Senior Citizen	0.1621	0.3686	0	0	0	0	1
Tenure (months)	32.37	24.56	0	9	29	55	72
Monthly Charges (USD)	64.76	30.09	18.25	35.50	70.35	89.85	118.75

The exploration step in the notebook remains intentionally lightweight limited to inspecting the head and tail of the dataframe, its shape, dtypes, and missing value counts rather than a full visual exploratory data analysis. The graphical outputs are produced at the model-evaluation stage; a panel of five confusion matrices, a combined ROC-curve comparison, and (new in this version) a grouped bar chart comparing accuracy, precision, recall, F1-score, balanced accuracy, and ROC-AUC across all five models, discussed in Sections 5 and 6.

3.3 Data Preprocessing Techniques

The data preprocessing pipeline follows the following sequence of techniques, including irrelevant-column removal, data-type conversion, missing-value handling, target encoding, categorical feature encoding, stratified dataset partitioning and feature standardization to prepare the Telco Customer Churn data for subsequent model training and evaluation.

- **Column removal:** The non-predictive `customerID` identifier column was dropped because it uniquely identifies customers and does not provide useful predictive information for churn classification.
- **Type coercion and imputation:** `TotalCharges` was converted from string to numeric using `pd.to_numeric(..., errors="coerce")`, which turns the blank entries into `NaN`; these were then filled with `0`, reflecting customers with no billing history yet (`tenure = 0`).
- **Target encoding:** The `Churn` label was mapped from the categorical values {"Yes", "No"} to the binary integers {1, 0}.This allows the classification algorithms to work with numerical class labels.
- **Categorical encoding:** All remaining categorical predictors (`gender`, `Partner`, `Dependents`, `PhoneService`, `MultipleLines`, `InternetService`, `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport`, `StreamingTV`, `StreamingMovies`, `Contract`, `PaperlessBilling`, `PaymentMethod`) were one-hot encoded via `pd.get_dummies(drop_first=True)`, expanding the 19 raw predictors into 30 numeric feature columns suitable for distance-based and gradient-based algorithms.
- **Train/test partitioning:** The data was split 80% / 20% into training (5,634 samples) and test (1,409 samples) subsets using `train_test_split` with `stratify=y` and `random_state=42`, preserving the 26.5% churn class proportion in both partitions (374 churners and 1,035 non-churners in the test set).Stratification preserves the class distribution across the two partitions.It was used so that the proportion of churned and non-churned customers remained approximately consistent between the training and testing sets.
- **Feature standardization:** A `StandardScaler` was fit on the training features only and then applied to transform both the training and test features, giving every feature zero mean and unit variance essential for the distance-based Custom KNN classifier and beneficial for the SVM and MLP models.This prevents information from the test set from influencing the scaling parameters and produces standardized features for the subsequent classifiers.

4. Machine learning techniques for predicting customer churn

Machine learning is a useful approach for predicting customer churn because it can learn patterns from previous customer data and use those patterns to predict whether a customer is likely to leave or stay. In the telecom industry, customer churn is generally treated as a binary classification problem, where the outcome has two possible values: Yes (the customer churns) and No (the customer stays). Different machine learning techniques can be applied to this problem, and their performance can be compared to find the most suitable model.

In this project, several classification algorithms are used to predict customer churn. Each algorithm works differently and has its own strengths and limitations. Using multiple techniques is helpful because one model may perform better than another depending on the characteristics of the dataset.

4.1. Decision Tree Classifier (DT) Model

The Decision Tree Classifier (DT) is a supervised machine learning algorithm that can be used to predict whether a customer will churn or not churn. It works by making a series of simple decisions based on the features in the dataset. The model starts from a root node and divides the data into smaller groups based on different conditions. This process continues until it reaches a leaf node, where the final prediction is made. The Gini impurity is calculated using the following equation:

$$\mathrm{Gini}(S) = 1 - \sum_{i=1}^{C} p_i^2$$

4.2. K-nearest neighbors (KNN) model

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm that can be used for classification problems such as predicting whether a customer will churn or stay. Unlike some other machine learning algorithms, KNN does not build a complex model during training. Instead, it stores the training data and makes a prediction by comparing a new customer with customers who have similar characteristics.The class with the highest frequency among these neighbors is assigned to the new observation.A commonly used distance measure is Euclidean distance:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

where x and y represent two observations and n represents the number of features.

4.3. Naive Bayes' Theorem

Naive Bayes is a simple and popular machine learning technique used for classification problems. It is based on Bayes' Theorem, which helps us find the probability of an event based on the information that is already available. In this project, Naive Bayes can be used to predict whether a customer is likely to churn or stay. Bayes' theorem is expressed as:

$$P(C \mid X) = \frac{P(X \mid C)P(C)}{P(X)}$$

where:

- $P(C \mid X)$ is the posterior probability of class C given the feature vector X.
- $P(X \mid C)$ is the likelihood of observing X given class C.
- $P(C)$ is the prior probability of class C.
- $P(X)$ is the probability of observing the feature vector.

4.4. Multilayer perceptron (MLP) model

Multilayer Perceptron (MLP) is a type of artificial neural network that is used for classification and prediction. It is designed to learn patterns from data and use those patterns to make predictions. In this project, MLP is used to predict whether a customer is likely to leave the telecom company or continue using its services.

MLP consists of an input layer, one or more hidden layers, and an output layer. The input layer receives customer information, while the hidden layers learn patterns from the data. The output layer gives the final prediction as Churn = Yes or Churn = No. The main formula used in a MLP is::

$$z = \sum_{i=1}^{n} w_i x_i + b$$

and the activated output is:

$$a = f(z)$$

where w_i represents the weight associated with input x_i, b represents the bias, and f represents the activation function.

4.5. Support Vector Machine (SVM) Model

Support Vector Machine (SVM) is a machine learning algorithm used for classification problems. In this project, it can be used to predict whether a customer is likely to churn or stay. The basic idea of SVM is to find the best boundary that separates the two groups of customers as clearly as possible.For a linearly separable dataset, the decision function can be represented as:

$$w^T x + b = 0$$

where w represents the weight vector and b represents the bias.

5. Model evaluation metrics

After training the different machine learning models, it is important to check how well they are performing. For this project, model evaluation metrics are used to compare the predicted results with the actual customer churn results. Since the project is about predicting whether a customer will churn or stay, several metrics are used instead of depending only on accuracy.A confusion matrix is used to understand how correctly a machine learning model classifies the data. In this project, the model predicts whether a customer will churn or not churn.

The confusion matrix has four possible results:

-True Negative (TN): The customer actually did not churn, and the model also predicted No churn.

-False Positive (FP): The customer actually did not churn, but the model predicted Churn.

-False Negative (FN): The customer actually churned, but the model predicted No churn.

-True Positive (TP): The customer actually churned, and the model correctly predicted Churn.

Accuracy: The model's overall accuracy is calculated as the ratio of correct predictions to the total number of predictions.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{(\text{TP} + \text{TN}) + (\text{FP} + \text{FN})}$$

It considers both correctly predicted income classes and gives a general indication of how well the model performs on the complete dataset.

Precision : Precision is used to determine how many of the records predicted as belonging to a particular class are actually members of that class. A higher precision indicates that the model makes fewer incorrect positive predictions. Its formula is:

$$\text{Precision} = \frac{\text{TP}}{(\text{TP} + \text{FP})}$$

Sensitivity (Recall): Recall measures how many of the actual records belonging to a class are correctly identified by the model. This measure is useful for seeing whether the model is failing to identify some of the actual cases.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1-score : F1 -score considers both precision and recall and gives a single value that represents their balance. It is useful when both types of errors are important in evaluating the classification model.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Balanced Accuracy : Balanced Accuracy considers the performance of both income classes and is useful when the classes are not equally distributed.

$$\text{Balanced Accuracy} = \frac{\text{Sensitivity} + \text{Specificity}}{2}$$

ROC-AUC: ROC-AUC is a performance metric used to measure how well a model distinguishes between customers who churn and those who stayds. The formulas are:

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

$$\mathrm{FNR} = \frac{\mathrm{FN}}{\mathrm{FN} + \mathrm{TP}}$$

6. Result

Table 5 reports the full set of evaluation metrics obtained for each of the five classifiers on the 1,409-sample held-out test set. The tuned MLP neural network achieved the highest F1-score (60.22%), ROC-AUC (83.90%), MCC (46.23%), Cohen's kappa (46.21%), balanced accuracy (72.83%) and G-mean (71.52%) of all five models, making it the strongest single model by most balanced criteria. The tuned Decision Tree achieved the highest raw accuracy (79.42%), the highest specificity (88.60%) and the lowest false positive rate (11.40%), reflecting its shallow, well-regularized structure after grid search. The tuned SVM with a linear kernel performed competitively across the board (78.64% accuracy, 82.38% ROC-AUC) without being the top performer on any single metric, while achieving the lowest log loss (0.4431) of the three tuned models, indicating well-calibrated probability estimates.

As in the baseline workflow, Gaussian Naïve Bayes recorded the lowest accuracy (65.58%) and by far the highest false-positive rate (42.03%), but the highest sensitivity/recall (86.63%), the lowest false-negative rate (13.37%), and the highest Fowlkes-Mallows index (60.81%). Its log loss of 2.7299 is dramatically higher than every other model's, reflecting the well-known tendency of Gaussian Naïve Bayes to produce overconfident, poorly calibrated probability estimates when its conditional-independence assumption is violated, as is the case here given the many correlated one-hot-encoded service features. The custom KNN implementation performed the weakest of the four non-Naïve-Bayes models on nearly every metric in this tuned comparison, in contrast to the baseline run, where it had outperformed the untuned Decision Tree; this reversal is consistent with the Decision Tree benefiting substantially more from grid-search tuning than a fixed-k KNN model, which had no hyperparameters adjusted.

Table 5: Performance results of the machine learning classifiers on the stratified test set

Metric	Custom KNN	Decision Tree	SVM	Naïve Bayes	MLP Neural Net
Accuracy (%)	76.93	79.42	78.64	65.58	79.28
Precision (%)	56.94	63.12	61.30	42.69	61.39
Sensitivity / Recall (%)	53.74	54.01	52.94	86.63	59.09
Specificity (%)	85.31	88.60	87.92	57.97	86.57
F1-score (%)	55.30	58.21	56.81	57.19	60.22
Balanced Accuracy (%)	69.53	71.30	70.43	72.30	72.83
ROC-AUC	0.8082	0.8267	0.8238	0.8092	0.8390
MCC	0.3980	0.4491	0.4293	0.3951	0.4623
Log Loss	0.6132	0.4552	0.4431	2.7299	0.4261

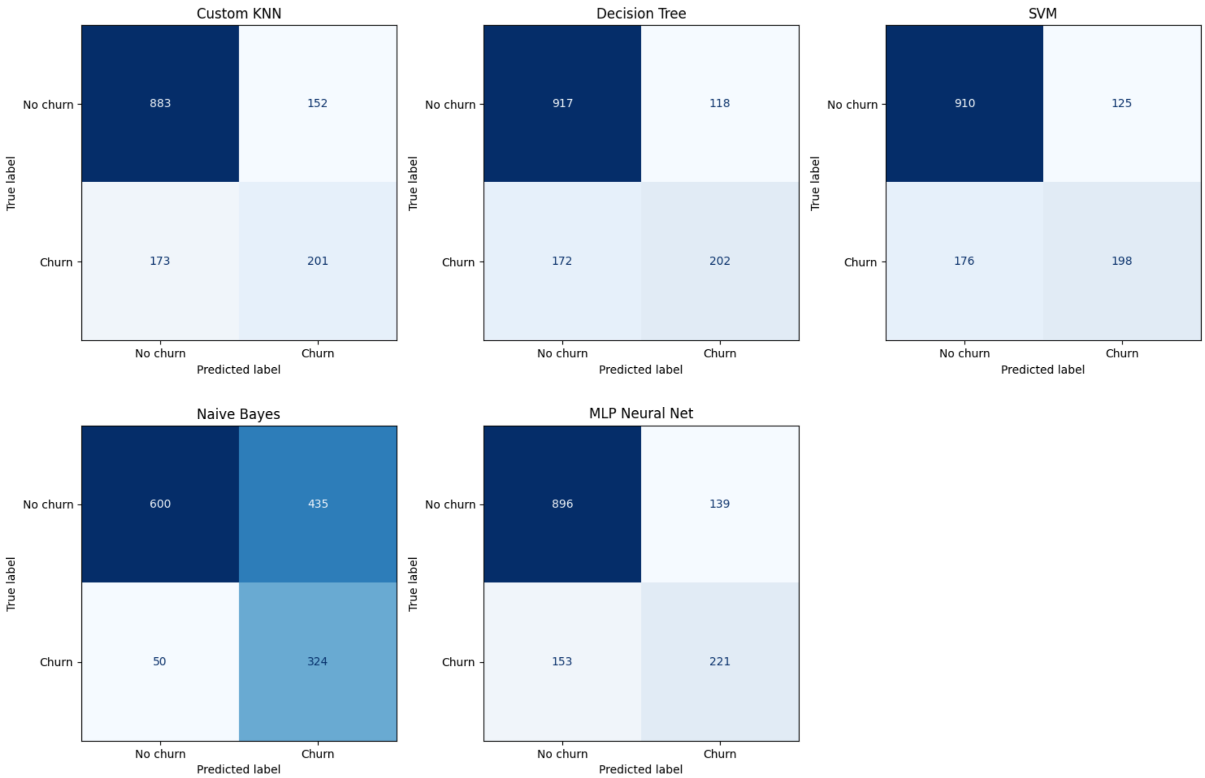

Figure 1: Confusion matrices generated by the supplied implementation.

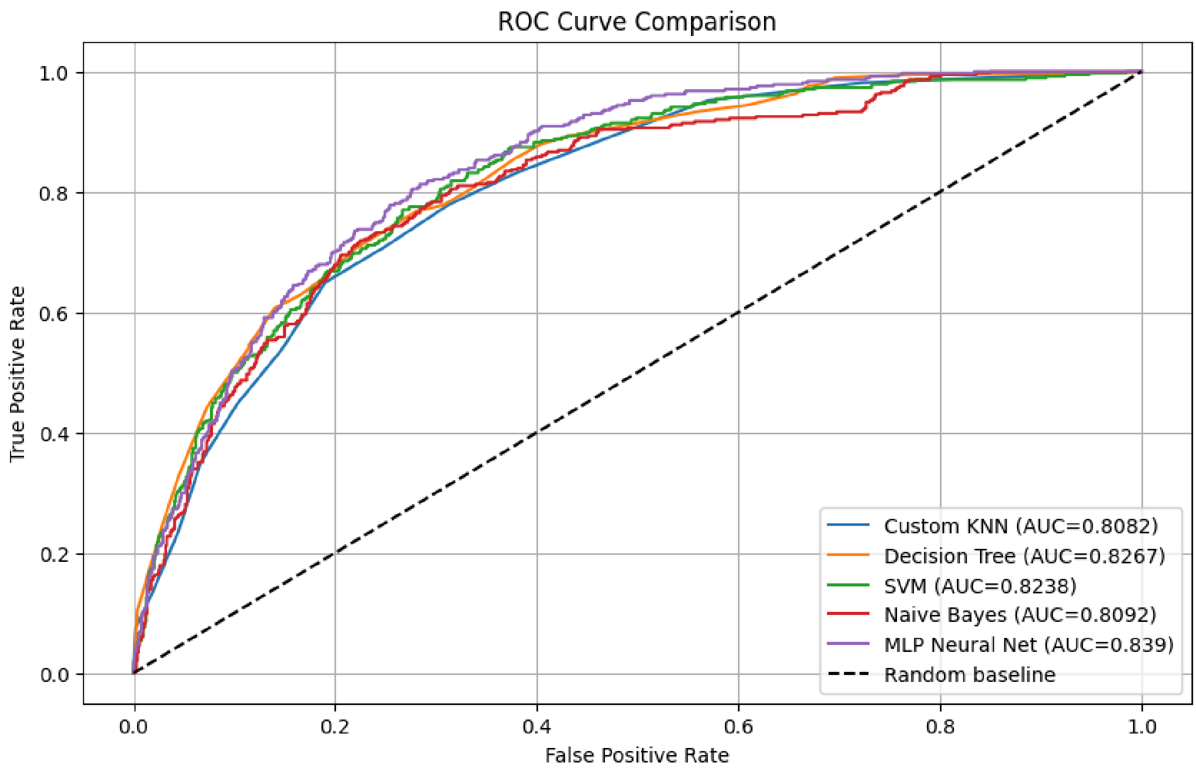

Figure 2: ROC curve comparison for the five classifiers.

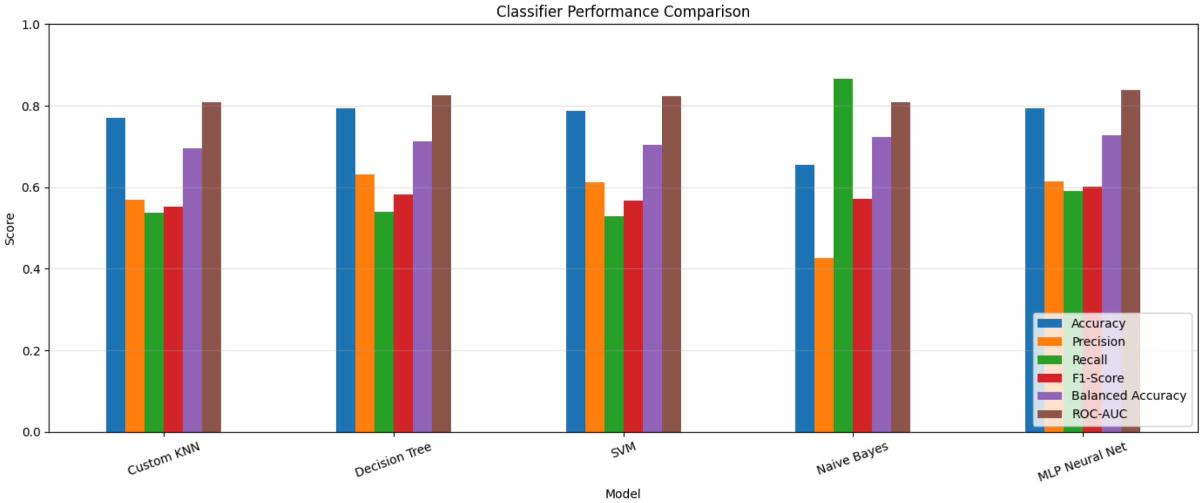

Figure 3: Comparative performance across the principal evaluation metrics.

From the confusion matrices, most of the models were able to correctly identify customers who did not churn. The Decision Tree correctly classified the highest number of non-churn customers (917), while the Naive Bayes model correctly identified the highest number of churn customers (324). However, Naive Bayes also produced more false predictions, so its overall accuracy was lower.

Looking at the performance comparison, the Decision Tree, SVM, and MLP Neural Network achieved similar accuracy, around 79%. The MLP Neural Network performed slightly better overall because it had the highest F1-score (0.60) and ROC-AUC (0.839). This means it was better at maintaining a balance between correctly identifying churn customers and avoiding incorrect predictions.

The Naive Bayes model had the highest recall (0.87). In simple terms, it was able to identify most of the customers who actually churned. However, its precision was quite low because it also predicted many non-churn customers as churn customers.

The ROC curve also supports these findings. The MLP Neural Network had the highest AUC of 0.839, followed by the Decision Tree (0.8267) and SVM (0.8238). Therefore, based on the overall results, MLP Neural Network can be considered the best-performing model, while the Decision Tree is also a strong and reliable model.

Table 6 Comparison of training and test accuracy for overfitting analysis

Model	Training Accuracy	Test Accuracy	Accuracy Gap
Custom KNN	0.8126	0.7693	0.0432
Decision Tree	0.8021	0.7942	0.0079
SVM	0.8001	0.7864	0.0138
Naive Bayes	0.6654	0.6558	0.0096
MLP Neural Net	0.8186	0.7928	0.0258

Overfitting Analysis: The training and test accuracies were compared to assess the generalization performance of the five classifiers. The Decision Tree showed the smallest accuracy gap of 0.0079, followed by Naïve Bayes with a gap of 0.0096, indicating relatively similar performance between training and unseen test data. The SVM and MLP Neural Network showed moderate gaps of 0.0138 and 0.0258, respectively. Custom KNN had the largest gap of 0.0432, suggesting comparatively greater sensitivity to the training data. Overall, all models exhibited accuracy gaps below 5 percentage points, suggesting that substantial overfitting was not observed based on this measure. The results indicate that the evaluated models were able to generalize reasonably well to the held-out test set.

7. Conclusion

This report evaluated five machine learning classifiers: a from-scratch K-Nearest Neighbors implementation, a Decision Tree, a tuned Support Vector Machine, a Gaussian Naive Bayes classifier and a tuned Multilayer Perceptron. All of these were used to predict customer churn on the Telco Customer Churn dataset. After a preprocessing pipeline that handled values, encoded categorical variables performed stratified splitting and standardized features hyperparameters for Decision Tree Support Vector Machine and Multilayer Perceptron were optimized using a grid search that maximised the F1-score under 3-fold stratified cross-validation.

The Multilayer Perceptron delivered the overall balance of predictive performance leading on F1-score (0.6022) balanced accuracy (0.7283) ROC-AUC (0.8390) and MCC (0.4623). Decision Tree was an more interpretable runner-up. Gaussian Naive Bayes while overall and poorly calibrated (log loss = 2.7299) uniquely offered the highest churner-detection rate (recall = 0.8663). This demonstrates that model selection for churn prediction should be guided by the retention program's tolerance for positives rather than by a single aggregate metric.

The comparison of training and held-out test accuracy showed relatively small generalization gaps across all five models. The largest gap was observed for Custom KNN at 4.32 percentage points, while the Decision Tree showed the smallest gap at 0.79 percentage points. Overall, substantial overfitting was not observed based on the training-test accuracy gap alone. Future work could extend this comparison by: (i) applying class-balancing techniques such as SMOTE or class-weighted loss functions to address the 26.5% positive-class prevalence; (ii) evaluating stronger ensemble methods such as Random Forest, XGBoost and LightGBM alongside the models studied here; (iii) tuning the Custom KNN implementation's neighbourhood size and distance weighting or replacing raw one-hot features with a lower-dimensional embedding to close its accuracy gap; (iv) recalibrating Naive Bayes' probability outputs to make its high recall operationally trustworthy for risk scoring; and (v) performing threshold optimisation on the strongest models Multilayer Perceptron and Decision Tree to align the precision–recall trade-off, with the specific cost structure of the retention program in which the model would be deployed.

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